# Type I / Type II + α / β / basic power

A1S2: SLIDES 1-51 ONLY

## 1 RECOGNISE IT

- Asked to classify a wrong decision or compare α, β and power.
- Need qualitative effects on β and power.
- Truth of $H_0$ and the decision are both specified.

## 2 DECISION / METHOD

Use a 2x2 truth/decision grid:
- Reject true $H_0$ -> Type I.
- Do not reject false $H_0$ -> Type II.
- Correct rejection of false $H_0$ has probability power.

## 3 FORMULA

```
P(Type I)=α
P(Type II)=β
power=1-β
matching confidence coefficient=1-α
```

## 4 FORMULA SHEET VS WHAT I KNOW

Gives: no error-definition formulas.
Still know - ALL ELSE FIXED: β rises when the true effect is smaller, α falls, variance rises, or n falls; power moves opposite β.

## 5 EXCEL ROUTE

No required calculation route here. A decision label can use:
`=IF(p_value<alpha,"Reject H0","Do not reject H0")`

## 6 TINY WORKED EXAMPLE

A real effect exists, but the test does not reject $H_0$: that is Type II. If β=.20, basic power is .80.

## 7 TOP EXAM TRAPS

- Type I = 'accepting false $H_0$' (wrong).
- Saying α+β=1.
- Confusing matching confidence coefficient 1-α with power 1-β.
- Omitting the ALL ELSE FIXED qualification.

## 8 SAY ALOUD

Type I rejects a true $H_0$; Type II misses a false $H_0$; power is the chance of detecting a false $H_0$.

## 9 EXIT CHECK

If n increases with all else fixed, what generally happens to β and power?

**ONE-GLANCE VISUAL ROUTE**

| | $H_0$ TRUE | $H_0$ FALSE |
|---|---|---|
| **REJECT $H_0$** | **TYPE I probability α** | **CORRECT power = 1-β** |
| **DO NOT REJECT $H_0$** | **CORRECT probability 1-α** | **TYPE II probability β** |